

EXPT.
NO. 1

NAME

Load Test on a single-phase (1- Φ) Transformer.

Page No.: 01

YOUVA

Date: 05/09/24

◦ Aim:

- To determine the efficiency and regulation of a single-phase transformer by conducting direct load test.

◦ Apparatus:

	Equipment	Range	Quantity
1.	1- Φ Transformer	230V/115V	1
2.	1- Φ Autotransformer/Variac	0-260V	1
3.	Wattmeter	75/150/300V, 5/10A 300/600V, 5/10A	2
4.	Voltmeter (AC/MI)	0-300V 0-150V	2
5.	Ammeter (AC/MI)	0-10A 0-5A	2
6.	Resistive Load	2kW, 230V	1
7.	Connecting wires	-	Required

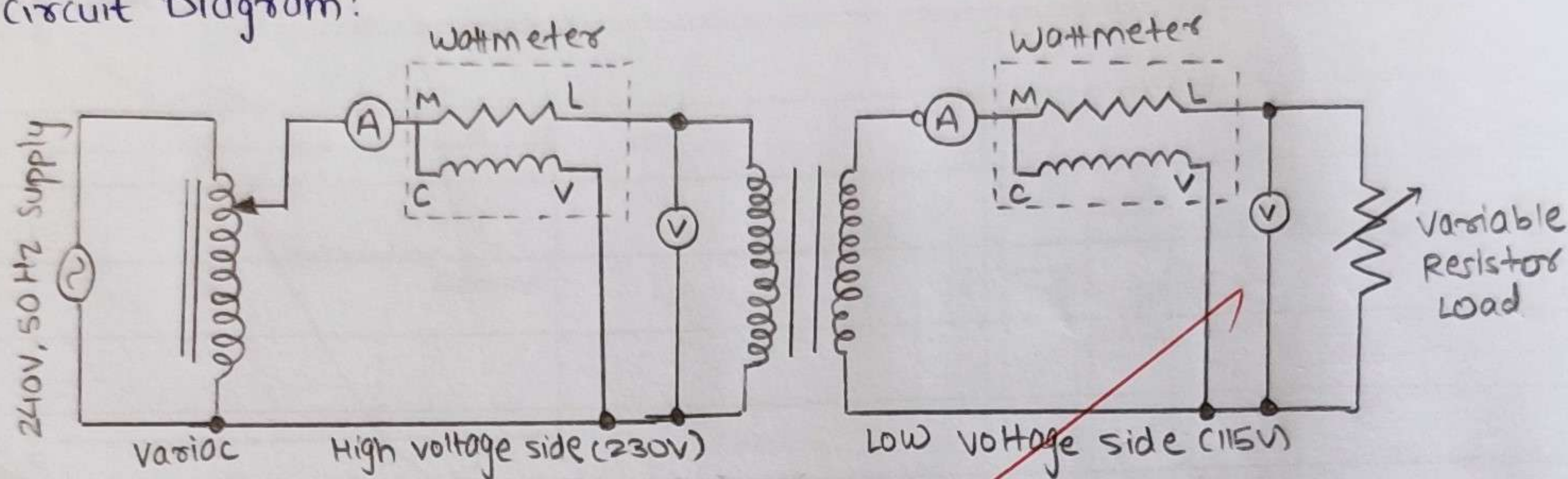
◦ Result:

Direct Load Test is performed on a 1- Φ transformer.
Efficiency and Regulation is plotted w.r.t load current.

~~Large
value~~

Teacher's Signature: _____

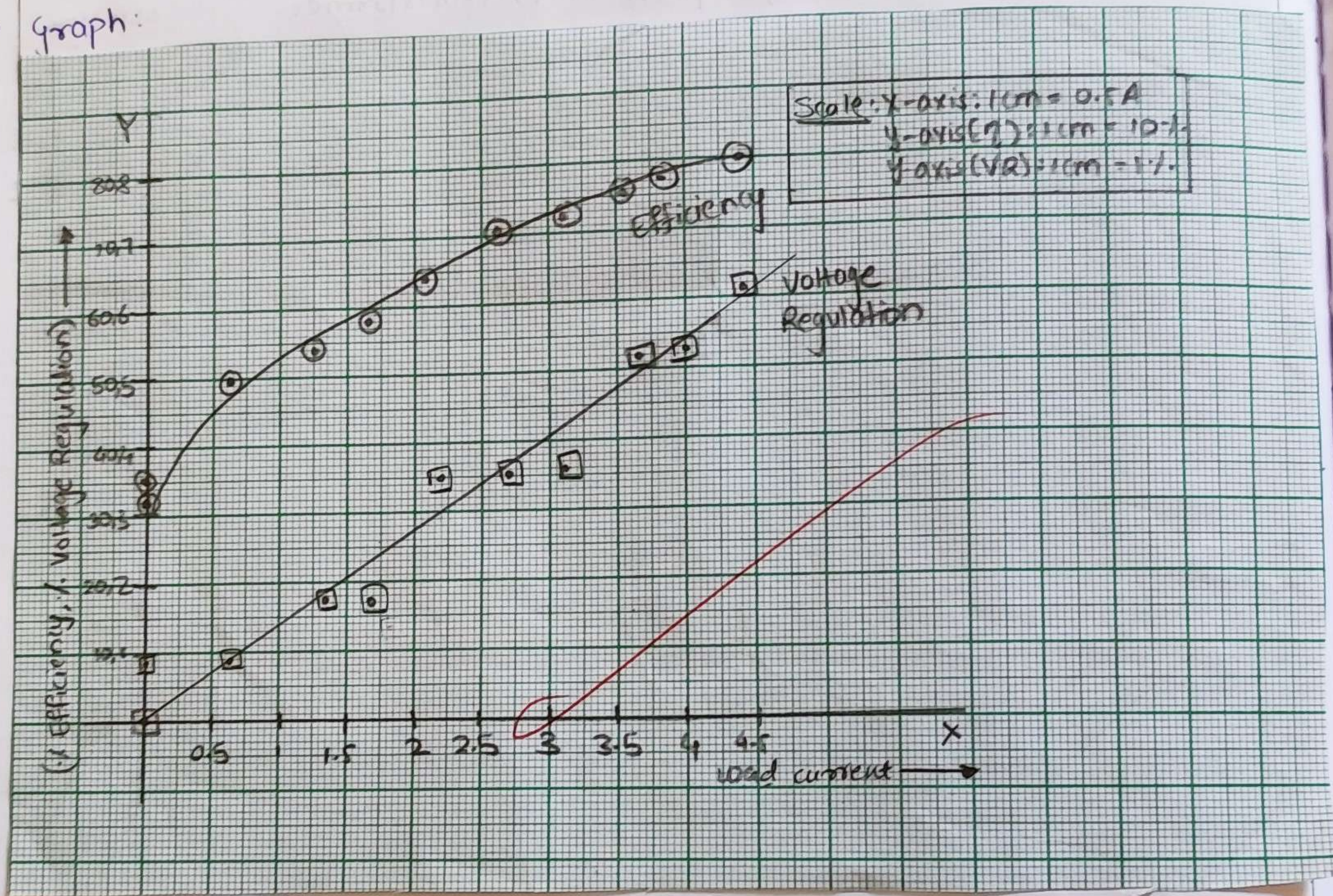
Circuit Diagram:



Observation Table:

Primary Voltage $V_1 (V)$	Primary Current $I_1 (A)$	Primary Power $P_1 (W)$	Secondary Voltage $V_2 (V)$	Secondary Current $I_2 (A)$	Secondary Power $P_2 (W)$	Efficiency $\eta = \frac{P_2}{P_1} \times 100$	Voltage Regulation (%)
228	0	$15 \times 4 = 60$	114	0	$10 \times 2 = 20$	33.33	0
228	0.33	$35 \times 4 = 140$	113	0	$25 \times 2 = 50$	35.71	0.88
228	0.48	$55 \times 4 = 220$	113	0.66	$55 \times 2 = 110$	50	0.88
228	0.66	$70 \times 4 = 280$	112	1.32	$75 \times 2 = 150$	54.57	1.78
228	1.05	$85 \times 4 = 340$	112	1.66	$95 \times 2 = 190$	58.88	1.78
228	1.25	$95 \times 4 = 380$	110	2.2	$122 \times 2 = 244$	64.21	3.5
228	1.50	$105 \times 4 = 420$	110	2.7	$150 \times 2 = 300$	71.43	3.5
228	1.75	$120 \times 4 = 480$	110	3.1	$175 \times 2 = 350$	72.916	3.5
228	1.95	$130 \times 4 = 520$	108	3.6	$200 \times 2 = 400$	76.92	5.26
228	2.15	$140 \times 4 = 560$	108	4	$220 \times 2 = 440$	78.57	5.26
228	2.35	$150 \times 4 = 600$	107	4.4	$240 \times 2 = 480$	80	6.14
228							

Graph:



% Efficiency and % Voltage Regulation vs. Load current graph.

o Theory:

- A transformer operates on the principle of "Electromagnetic Induction". It is a two winding equipment, which has "ELECTRICALLY ISOLATED, MAGNETICALLY COUPLED" windings. The windings are placed on a supporting structure called the CORE which is made up of material of High permeability.

- A transformer can be operated to either increase or decrease the voltage on the secondary side depending on transformation ratio 'K'.

$$K = \frac{\text{Secondary side voltage}}{\text{Primary side voltage}} = \frac{\text{No. of turns in 2}^\circ \text{ side}}{\text{No. of turns in 1}^\circ \text{ side}} = \frac{\text{1}^\circ \text{ side current}}{\text{2}^\circ \text{ side current}}$$

$$\text{Hence, } K = \frac{V_2}{V_1} = \frac{N_2}{N_1} = \frac{i_1}{i_2}$$

If value of K is,

- i) $> 1 \Rightarrow$ Transformer is step-up transformer.
- ii) $< 1 \Rightarrow$ Transformer is step-down transformer.
- iii) $= 1 \Rightarrow$ Transformer is isolator transformer.

- Efficiency and regulation of a transformer at any desired load condⁿ and power factor can be found either by directly loading it.

$$\therefore \text{Efficiency } (\eta) = \frac{\text{Output Power}}{\text{Input Power}} \times 100$$

Under no-load condition (when secondary circuit is open) and $i_2 = 0$, voltmeter reading on secondary side gives no-load secondary terminal voltage, V_{02} . Hence,

$$\therefore \text{Voltage Regulation} = \frac{V_{02} - V_2}{V_{02}}$$

◦ Procedure:

- Connections are to be made as per the circuit diagram.
- Double pole single throw (DPST) switch is closed.
- Under no load condition, ammeter, voltmeter, and wattmeter readings on both primary and secondary side are noted down.
- The load is increased in suitable intervals and for each load interval, corresponding reading of voltmeter, ammeter, and wattmeter on both primary and secondary sides are noted down.
- The experiment is repeated until the rated current of the transformer (take the minimum rated current of the transformer primary side) has been reached.
- The load is reduced to bring the transformer to the no-load condition. The auto-transformer is brought to its minimum position and then the DPST switch is opened.
- Calculate the voltage regulation and % efficiency at each load and plot the graphs.